What Is an Alarm Event?

The simple structure behind a reported alarm

A change in a known device's condition is one kind of event. A report may describe that change, a restoration, a continuing condition, or a status observation. The device remains identifiable while the system communicates what it knows about the condition.

Core sentence

This system reports that this device, of this type, at this location, has changed to this state.

The structure of an event

DEVICE = (ID, TYPE, LOCATION)
STATE = Normal | Trouble | Active
EVENT = ΔSTATE × DEVICE × TIME

These equations describe conceptual relationships, not arithmetic operations. The three states are simplified human-facing categories, not the six SAFER reporting STATUS values.

An event is not the device itself. A device is identified within its reporting system by its ID and source context; TYPE describes what it is, and LOCATION describes where it belongs. A report may describe a new change, a restoration, or a previously reported condition.

What describes the device

- ID — Which point is involved
- TYPE — What kind of point it is
- LOCATION — Where it belongs

What the state means to people

- Normal — A monitored condition reported or otherwise established as normal.
- Trouble — Investigate; operation may be impaired
- Active — Act according to the applicable plan

Mirror effect

A panel message, printer line, database record, floor-plan symbol, or SAFER JSON report may look different while referring to the same point and condition. The format changes; the underlying information roles remain recognizable.

Device	State	Meaning for people
Smoke Detector, Classroom 4 West	Normal	Reported or otherwise established as normal.
Smoke Detector, Classroom 4 West	Trouble	Investigate; operation may be impaired
Smoke Detector, Classroom 4 West	Active	Act according to the applicable plan

Takeaway: ID, TYPE, and LOCATION describe the known point. STATE describes its condition. A report may communicate a change, restoration, repetition, or status observation.